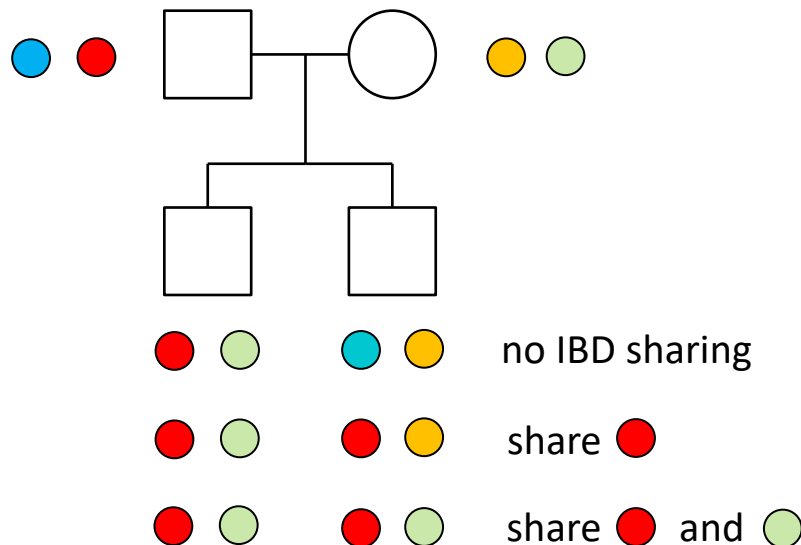


# IBD coefficients

- Summary so far:
  - Two individuals are related if they can have IBD alleles
  - Their kinship coefficient measures the amount of IBD sharing
- Natural generalisation:
  - How *many* alleles are IBD in each locus?



## Definition

- $\kappa_0 = Pr(0 \text{ alleles IBD})$
- $\kappa_1 = Pr(1 \text{ alleles IBD})$
- $\kappa_2 = Pr(2 \text{ alleles IBD})$

(at random autosomal locus)